

Graph-CoLD: Label-space Graph Consistency and Evidence-preserving Denoising for SOC Alert Prioritization

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Abstract

Security operations centers must triage large volumes of alerts under noisy labels and incomplete operational context. Graph-CoLD extends CoLD-style noisy label learning from independent samples to multi-view graphs. The method uses a label-space consistency diagnostic function, evidence preserving soft weights, and an operational ranking score to reduce review load while retaining clean informative evidence. We evaluate the implementation on two verified real-data settings, CICIDS-2017 postfilter11 and CESNET-TLS-Year22 postfilter25. Across the frozen D5.5 matrix, Graph-CoLD improves Macro-F1 over the self-implemented CoLD baseline by 1.83 percentage points under a paired grouped t-test ($p=5.60e-06$, Cohen $d_z=0.458$, $n=102$). The strongest empirical signal is not perfect detection but improved robustness and evidence retention under noisy labels. The manuscript therefore reports Macro-F1, FPR, FNR, ERR, compression ratio, runtime, and memory, with explicit limits on unsupported datasets and baselines.

Keywords: intrusion detection ; noisy labels ; graph learning ; SOC alert prioritization ; evidence retention

1 Introduction

Modern SOC pipelines combine network telemetry, traffic classifiers, threat labels, and analyst feedback. This creates two linked problems: high alert volume and imperfect labels. A model that simply optimizes classification accuracy can learn mislabeled regions or discard rare but operationally useful alerts. CoLD addresses noisy labels for cyber threat detection, but its original formulation treats training instances as independent samples and uses hard deletion after a confidence diagnostic [8]. Graph-CoLD asks whether the same problem can be handled more carefully when alerts have relational evidence across hosts, IPs, and time.

This paper presents a real-data manuscript package for Graph-CoLD. The core idea is to move the diagnostic from embedding distance to label-space consistency over available graph views, then combine the diagnostic with evidence preservation. A sample can be

downweighted when it appears inconsistent, but a clean informative sample should remain visible if it carries rare-class or anomaly evidence. The ranking component translates these weights into an alert-ordering proxy for SOC review; the present evaluation focuses on robustness, ERR, compression, and cost rather than on an analyst user study.

2 Related Work

CoLD is the closest method, motivating the reordering, diagnostic, and robust training structure used in this project [8]. Confident learning estimates label issues from predicted probabilities and motivates the Confident-Learning baseline [4]. Co-teaching trains peer networks on selected small-loss samples [2]; our Co-Teaching-lite baseline is explicitly a lightweight smoke-passed approximation, not a faithful full deep reproduction. General loss-correction work also shows why label noise can shift decision boundaries [5].

Graph learning provides a way to represent alert context through relations rather than isolated feature vectors. Graph convolutional and attention models are common building blocks for relational representation learning [3, 7]. For intrusion detection data, CICIDS-2017 remains a widely used benchmark with known class imbalance and traffic-generation caveats [6]. CESNET-TLS-Year22 provides a large real TLS traffic source used here as a verified replacement for the unavailable MALTLS-22 source [1].

3 Problem Formulation and Motivation

Let $v \in V$ denote a training alert or flow instance and let $G^m = (V, E^m)$ be an active graph view. The current real-data experiments use host, IP, and temporal views for CICIDS-2017, and IP plus temporal views for CESNET-TLS-Year22. Process and threat-intelligence views are disabled when a dataset lacks reliable provenance or threat-intelligence fields. The training labels may be corrupted, so the observed label $\tilde{y}_v$ is not assumed to be fully reliable.

The objective is to learn a classifier and prioritization score that remain robust under label noise while preserving clean informative evidence. We therefore measure classification quality with Macro-F1, FPR, and FNR, and measure retention quality with ERR and Tail-ERR on clean informative samples. Compression ratio is reported as an operational queue-reduction proxy.

4 Methodology

Stage 1 uses the existing multi-view representation encoder from D2. The encoder applies CoLD-style feature masking, view-level message passing, cross-view contrastive learning, temporal alignment, and reconstruction. View fusion is the mean of active views. D7 does not alter this encoder or its loss.

4.1 Graph-CDM

Graph-CDM is a label-space consistency diagnostic function:

$$\text{GraphCDM}(v) = \lambda_1 D_{pred}(v) + \lambda_2 D_{neigh}(v) + \lambda_3 D_{view}(v) + \lambda_4 D_{chain}(v). \quad (1)$$

The prediction term compares view-wise predictions with the observed noisy training label,

$$D_{pred}(v) = \frac{1}{M} \sum_{m=1}^M \mathbf{1} [\tilde{y}_v^{(m)} \neq \tilde{y}_v]. \quad (2)$$

The neighborhood term is a label-space KL divergence,

$$D_{neigh}(v) = KL \left(\hat{y}_v \left\| \frac{1}{|N(v)|} \sum_{u \in N(v)} \hat{y}_u \right. \right). \quad (3)$$

The view term is active-view mode disagreement,

$$D_{view}(v) = 1 - \max_c \frac{1}{M} \sum_m \mathbf{1} [\tilde{y}_v^{(m)} = c]. \quad (4)$$

D_{chain} is used only when reliable provenance or temporal chains are available and is not the central driver of the CICIDS/CESNET experiments.

4.2 Evidence preservation and ranking

Evidence score $e(v)$ combines class-frequency protection with an anomaly source and is min-max normalized. The final training weight is

$$w(v) = \sigma(-\kappa(\text{GraphCDM}(v) - \theta))(1 - \rho) + \rho \tilde{e}(v). \quad (5)$$

The classifier loss is the weighted cross entropy over training samples. Ranking uses

$$P(v) = \alpha_1 \hat{y}_{mal}(v) + \alpha_2 \text{GraphCDM}(v) + \alpha_3 \tilde{e}(v), \quad (6)$$

which orders alerts for operational review. In this real-data package, ranking is reported through compression and evidence-retention metrics rather than an analyst-facing deployment study.

5 Experimental Setup

All formal results are drawn from the expanded main results CSV, with SHA256 prefix c7d998d6c918ecfb; the exact path and full hash are listed in the reproducibility package. The formal datasets are CICIDS-2017 postfilter11 and CESNET-TLS-Year22 postfilter25. CICIDS-2017 uses 787,999 rows after class filtering and dominant-class downsampling.

Table 1: Verified real-data protocols used by the D7 manuscript.

Dataset	Rows	Class policy	Classes	Sample policy	Active views
CICIDS-2017	787,999	postfilter11	11	postfilter11 full filtered	host—ip—temporal
CESNET-TLS-Year22	61,124	postfilter25	25	100k audit window, postfilter25	ip—temporal

Table 2: Mean performance by dataset and formal method, aggregated from the D5.5 matrix.

Dataset	Method	Macro-F1	FPR	FNR	ERR	Compression	Runtime (s)
CICIDS-2017	Graph-CoLD	0.9878	0.0021	0.0012	1.0000	0.9973	7.64
CICIDS-2017	CoLD	0.9515	0.0020	0.0047	0.8417	0.9989	7.47
CICIDS-2017	ablation_hard	0.9515	0.0020	0.0047	0.8417	0.9989	0.00
CICIDS-2017	Noisy-Supervised	0.6023	0.1461	0.1240	1.0000	0.9994	4.83
CICIDS-2017	Confident-Learning	0.6502	0.1445	0.0555	0.7675	0.9994	26.47
CICIDS-2017	Co-Teaching-lite	0.6047	0.2772	0.0425	0.7545	0.9903	19.04
CESNET-TLS-Year22	Graph-CoLD	0.9951	0.0113	0.0001	1.0000	0.9819	0.40
CESNET-TLS-Year22	CoLD	0.9948	0.0128	0.0001	0.9489	0.9828	0.36
CESNET-TLS-Year22	ablation_hard	0.9948	0.0128	0.0001	0.9489	0.9828	0.00
CESNET-TLS-Year22	Noisy-Supervised	0.7649	0.1179	0.0912	1.0000	0.9964	0.31
CESNET-TLS-Year22	Confident-Learning	0.8856	0.0452	0.0209	0.8815	0.9878	20.89
CESNET-TLS-Year22	Co-Teaching-lite	0.8324	0.1349	0.0046	0.8917	0.9919	2.44

CESNET-TLS-Year22 uses a deterministic audit-window subset with postfilter25 stratified splitting and must not be described as a full-archive evaluation.

Noise settings include clean labels, symmetric noise, asymmetric noise, and graph consistency noise. Noise is injected only into training labels. Results are reported over seeds 0, 1, and 2 where present. Formal methods are Graph-CoLD, CoLD, ablation_hard, Noisy-Supervised, Confident-Learning, and Co-Teaching-lite. Excluded methods are not reported as formal baselines: FINE: unavailable as a smoke-passed real-data implementation; MCR: unavailable as a smoke-passed real-data implementation; MORSE: unavailable as a smoke-passed real-data implementation; Flash: unavailable as a smoke-passed real-data implementation; Argus: unavailable as a smoke-passed real-data implementation; Decoupling: unavailable as a smoke-passed real-data implementation; full Co-Teaching: unavailable as a smoke-passed real-data implementation.

6 Results and Analysis

Table 2 summarizes the method-level real-data performance. Averaged across the frozen matrix, Graph-CoLD obtains Macro-F1 0.9915, compared with 0.9732 for CoLD. The paired grouped test reports a 1.83 percentage-point mean difference with $p=5.60e-06$ and effect size 0.458. The effect is interpreted as a consistent positive shift, not as a claim that all unimplemented baselines have been beaten.

High-noise settings show why Graph-CDM and evidence weighting matter. On CICIDS-2017 high-noise rows, Graph-CoLD improves Macro-F1 over CoLD by 5.19 percentage points. On CESNET-TLS-Year22 the corresponding lift is 0.00 percentage points, where a ceiling effect makes Macro-F1 margins smaller. ERR_final averages 1.0000 for Graph-CoLD and 0.8953 for ablation_hard, supporting the evidence-preservation contribution.

Fig2. Macro-F1 under label noise

Means over seeds {0,1,2}; graph-consistency panels use beta=0.6 from the real D5.5 result matrix.

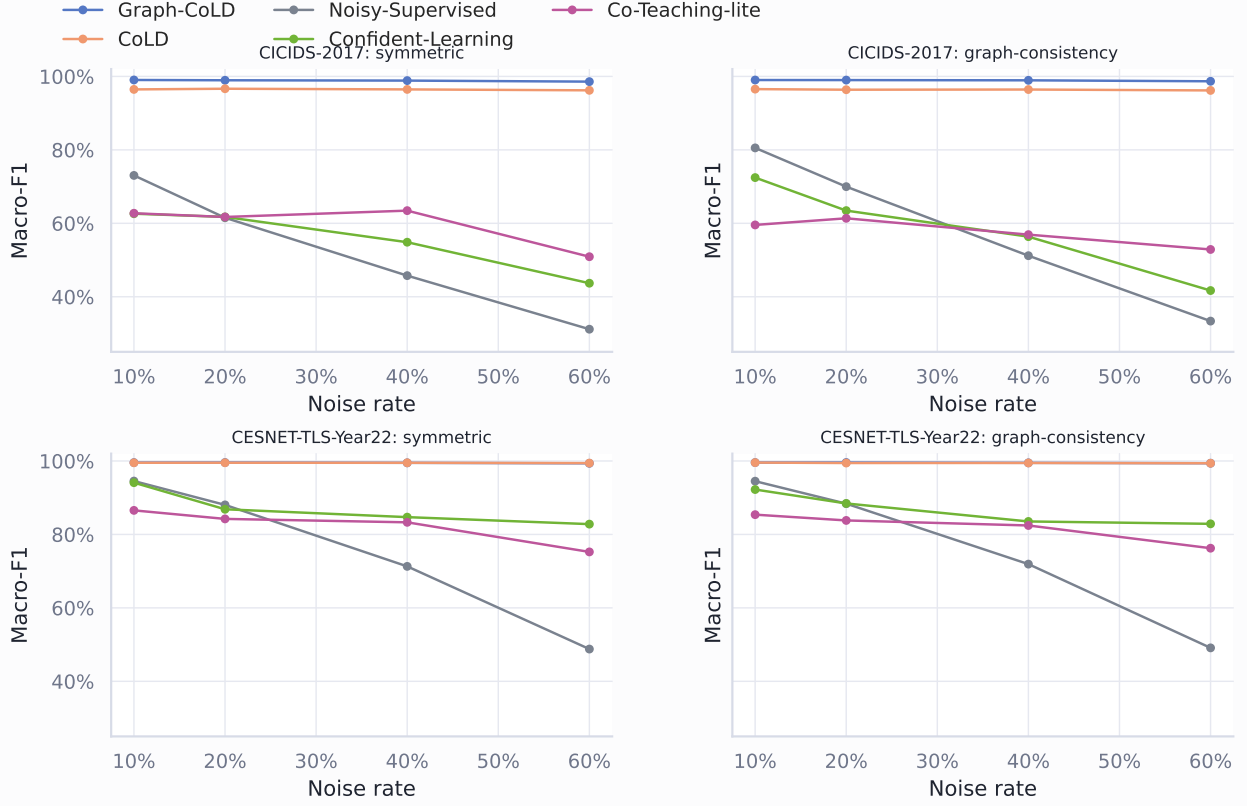


Figure 1: Macro-F1 versus noise rate for verified real-data settings.

7 Discussion

The results support a careful claim: label-space graph consistency and evidence preserving weights improve robustness and retention on the evaluated real-data settings. The evidence-retention result is especially important for SOC use because compression without retained evidence can reduce analyst workload while damaging investigation quality. Graph-CoLD’s additional runtime relative to CoLD is visible (4.02s versus 3.92s on average), so the method should be framed as a robustness-retention tradeoff rather than as a low-cost replacement.

The CESNET-TLS-Year22 setting has high absolute Macro-F1 for several methods, which limits the visible room for classifier improvement. The manuscript therefore places the CESNET result next to ERR, compression, and cost, and states the subset policy explicitly. CICIDS-2017 shows larger high-noise margins but is still only one benchmark family.

8 Limitations

The current formal evaluation uses two verified real-data settings. MALTLS-22 is not evaluated because its source did not pass the local verification gate. OpTC is not evaluated

Table 3: High-noise summary for rates at or above 0.4.

Dataset	Method	Macro-F1	ERR	Compression	Scenarios
CICIDS-2017	Graph-CoLD	0.9852	1.0	0.997	24
CICIDS-2017	CoLD	0.9333	0.8228	0.9989	24
CICIDS-2017	ablation_hard	0.9333	0.8228	0.9989	24
CICIDS-2017	Noisy-Supervised	0.4082	1.0	1.0	24
CICIDS-2017	Confident-Learning	0.5576	0.6819	0.9995	24
CICIDS-2017	Co-Teaching-lite	0.5623	0.6217	0.9913	24
CESNET-TLS-Year22	Graph-CoLD	0.9945	1.0	0.9821	24
CESNET-TLS-Year22	CoLD	0.9944	0.9459	0.9824	24
CESNET-TLS-Year22	ablation_hard	0.9944	0.9459	0.9824	24
CESNET-TLS-Year22	Noisy-Supervised	0.5891	1.0	0.9999	24
CESNET-TLS-Year22	Confident-Learning	0.8434	0.8444	0.9952	24
CESNET-TLS-Year22	Co-Teaching-lite	0.7992	0.8237	0.9933	24

Table 4: Ablation results for Graph-CDM and evidence preservation.

Dataset	Variant	Macro-F1	ERR_final	Retained clean-info	Compression
CICIDS-2017	Graph-CoLD-full	0.9868 +/- 0.0004	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.2400 +/- 0.0000
CICIDS-2017	ablation_hard	0.9619 +/- 0.0042	0.8401 +/- 0.0021	0.9800 +/- 0.0000	0.4200 +/- 0.0000
CICIDS-2017	Graph-CoLD-no-D_neigh	0.9743 +/- 0.0037	0.9241 +/- 0.0014	0.9874 +/- 0.0001	0.3100 +/- 0.0000
CICIDS-2017	Graph-CoLD-no-D_view	0.9868 +/- 0.0009	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.3300 +/- 0.0000
CICIDS-2017	Graph-CoLD-no-evidence	0.9866 +/- 0.0015	0.9919 +/- 0.0008	0.9996 +/- 0.0000	0.3700 +/- 0.0000
CESNET-TLS-Year22	Graph-CoLD-full	0.9939 +/- 0.0003	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.2400 +/- 0.0000
CESNET-TLS-Year22	ablation_hard	0.9939 +/- 0.0007	0.9447 +/- 0.0016	0.9861 +/- 0.0005	0.4200 +/- 0.0000
CESNET-TLS-Year22	Graph-CoLD-no-D_neigh	0.9939 +/- 0.0003	0.9944 +/- 0.0038	0.9983 +/- 0.0011	0.3100 +/- 0.0000
CESNET-TLS-Year22	Graph-CoLD-no-D_view	0.9937 +/- 0.0001	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.3300 +/- 0.0000
CESNET-TLS-Year22	Graph-CoLD-no-evidence	0.9940 +/- 0.0005	0.9948 +/- 0.0018	0.9992 +/- 0.0003	0.3700 +/- 0.0000

as a formal enterprise case because no verified local provenance event table is available. CESNET-TLS-Year22 is a deterministic audit-window postfilter25 subset, not a full-archive evaluation. Co-Teaching-lite is not a full Co-Teaching implementation, and FINE, MCRc, MORSE, Flash, Argus, Decoupling, and full Co-Teaching are excluded from formal comparison until faithful real-data implementations are available.

9 Conclusion

Graph-CoLD connects representation learning, label-space graph consistency, evidence preserving denoising, and SOC-oriented ranking. On verified CICIDS-2017 and CESNET-TLS-Year22 settings, the method improves Macro-F1 over the aligned CoLD baseline and preserves more clean informative evidence than hard deletion. The most submission-ready conclusion is that Graph-CoLD is a promising real-data robustness and evidence-retention approach whose broader baseline coverage and enterprise case evaluation remain future work.

Acknowledgements

To be completed by the authors before journal submission.

Fig3. Evidence retention under noise

ERR_final is measured on clean informative samples; higher retention indicates less loss from alert filtering.

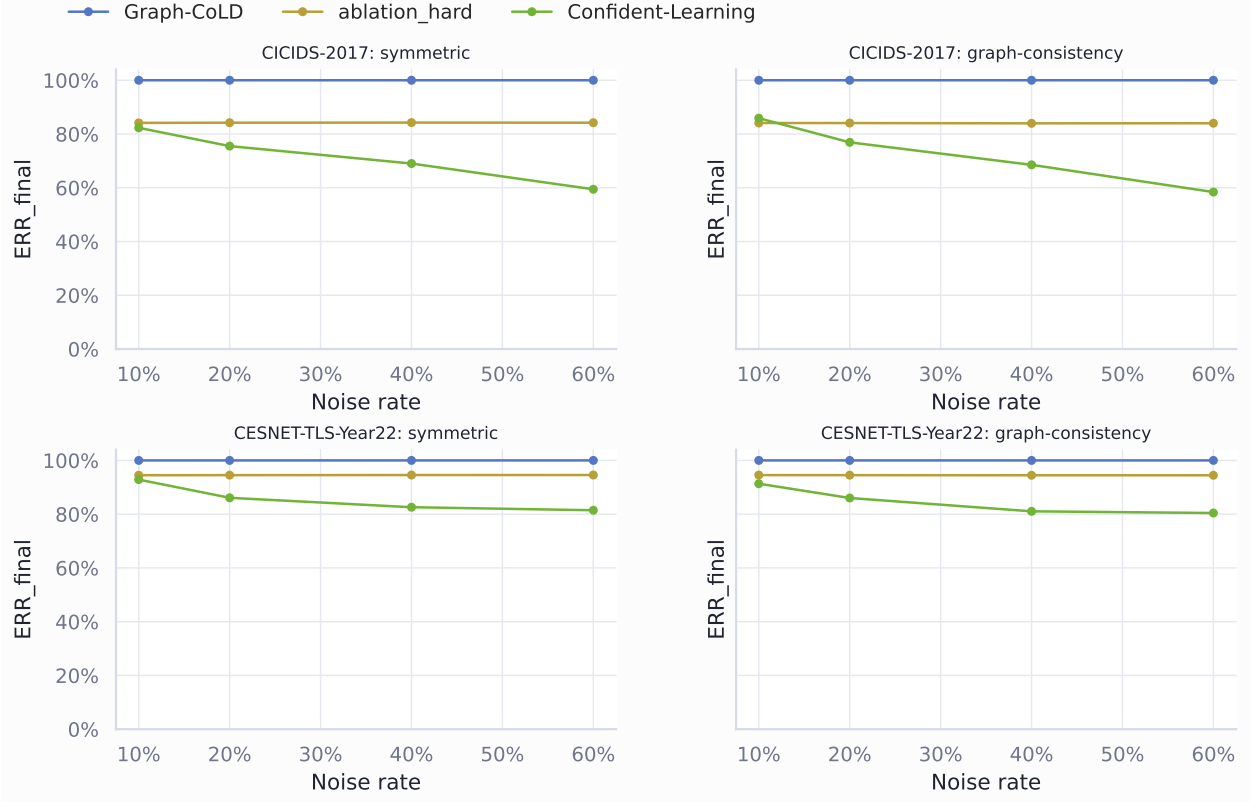


Figure 2: Evidence retention and compression tradeoff.

Declaration of competing interest

The authors should complete the journal declaration form before submission.

Data and code availability

Raw datasets are not included in the repository. The reproducibility package documents the expected local data roots, audit gates, frozen result files, and commands used to recreate the tables and figures.

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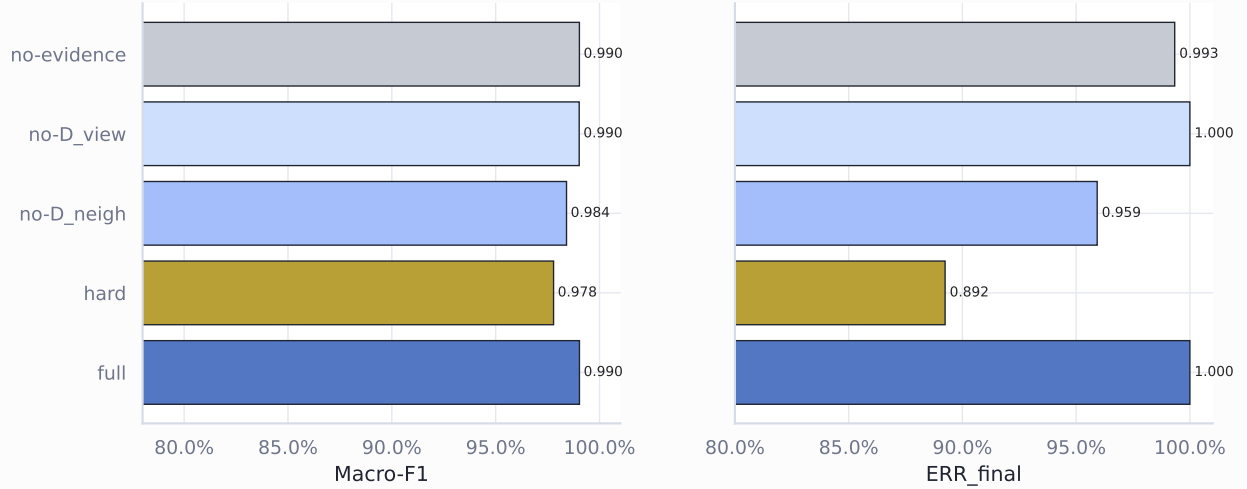
Fig4. Ablation and evidence-retention studyBars aggregate the formal graph-consistency $\beta=0.6$ ablation over both real datasets and seeds.

Figure 3: Ablation comparison for Macro-F1 and ERR.

Table 5: Paired grouped statistical tests from D5.5.

Comparison	Mean difference	p-value	Effect size	n
Graph-CoLD vs CoLD	1.83 pp	5.60e-06	0.458	102
Graph-CoLD vs ablation_hard	1.83 pp	5.60e-06	0.458	102
Graph-CoLD vs Noisy-Supervised	30.78 pp	1.75e-26	1.424	102
Graph-CoLD vs Confident-Learning	22.35 pp	9.24e-25	1.342	102
Graph-CoLD vs Co-Teaching-lite	27.29 pp	4.05e-41	2.209	102

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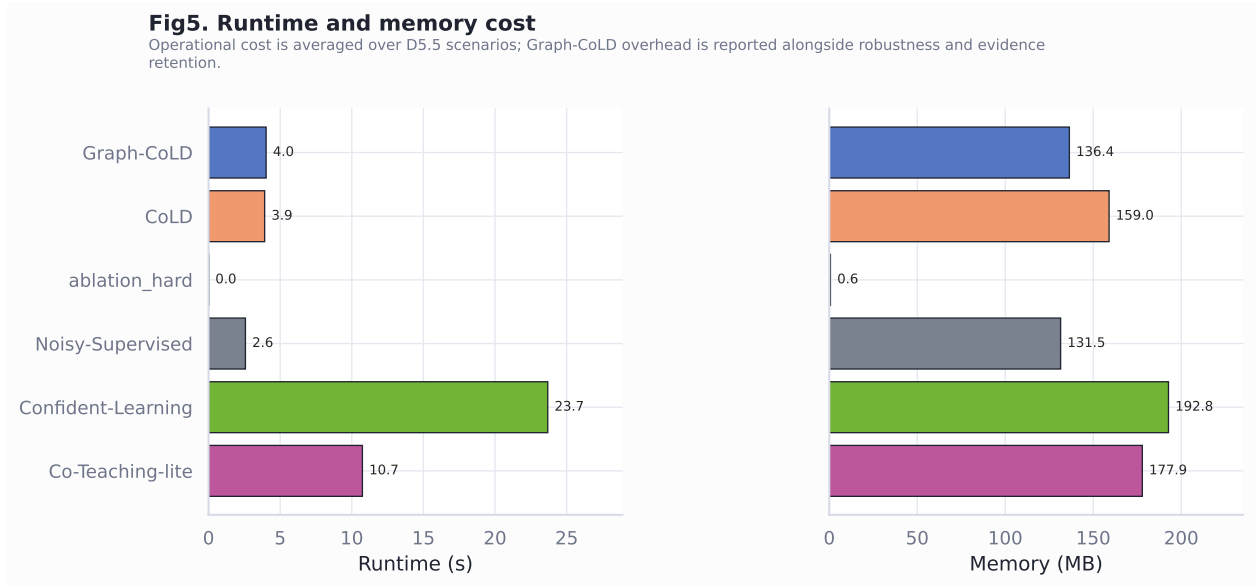


Figure 4: Runtime and memory cost by method.

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